

Grade 11 - Term 3

C1–C4 Practice Exercises and Solutions

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Evaluated criteria

- C1** Characterises continuous random variables by distinguishing them from discrete variables and interpreting their probability density function, cumulative distribution function, and support in economic contexts.
- C2** Applies uniform and triangular continuous distributions with constant or piecewise-linear probability density functions to calculate interval probabilities using geometric areas.
- C3** Applies circular-arc probability density functions to calculate interval probabilities using the geometry of circles and circular segments.
- C4** Applies the exponential distribution to constant-rate waiting-time situations and assesses its suitability.

1 Identifying Probability Distributions from Contexts [C1-5]

A variety of Colombian organizations use probability models to represent uncertain quantities and events. In each situation below, a random variable is defined from the information provided. The appropriate probability distribution must be identified from the characteristics of the situation.

Questions/Tasks.

- C1** For each context, characterize the random variable and identify the appropriate probability distribution. Briefly justify your choice based on the features described. [5]

Context 1. A payment platform monitors one scheduled electronic transfer. By the end of processing, it is either completed successfully or not completed. The variable X_1 records the outcome to help estimate verification and support needs.

Context 2. An agricultural technology company monitors the daily temperature variation inside a refrigerated storage unit. The variation X_2 ranges from -2°C to 2°C , with all values in this interval considered equally likely. The company uses this model to set an operating margin.

Context 3. A streaming platform sends the same offer to exactly 80 randomly selected eligible users. Each user either subscribes or does not. The variable X_3 counts the number of subscribers to estimate the server capacity needed for the promotion.

Context 4. A Bogotá parking company records the number of vehicle breakdowns during a fixed one-week period.

Comparable records show that breakdowns occur independently and without a predictable timetable. The variable X_4 counts the breakdowns to help schedule maintenance staff.

Context 5. An online marketplace studies the delivery time X_5 for a particular type of order. Delivery times range from 20 to 80 minutes, with deliveries around 45 minutes occurring most frequently. They become progressively less common toward either extreme. The company uses this model to estimate delivery times.

Context 6. A quality-control laboratory tests electronic components until the first defective component is found. Components are tested independently, and testing stops after the first defect. The variable X_6 records the number of components tested, including the defective one, to estimate inspection costs.

Context 7. A cloud-computing provider studies storage usage X_7 among small business customers. Usage ranges from 0 to 50 gigabytes and becomes steadily more common toward the upper limit, without reaching a peak before 50 GB. The provider uses this model to plan capacity.

Context 8. A retailer sends a promotional message to exactly 120 customers selected using the same procedure. Each customer either makes a purchase or does not. The variable X_8 counts the number of purchases to estimate the inventory needed for the campaign.

Context 9. A coffee exporter measures the moisture content X_9 of coffee bags under usual storage conditions. Measurements cluster around 11

Context 10. A food-delivery company records the number of customer cancellations during a fixed three-hour dinner period in one delivery zone. Comparable evenings show cancellations occurring independently without a recurring timetable. The variable X_{10} counts the cancellations to estimate customer-service needs.

C1 - Distribution identification: Solutions

[10]

The appropriate distribution for each context is determined by whether the variable is discrete or continuous and by the defining features of the situation:

Context	Type	Distribution	Characterization
1	Discrete	Bernoulli	There is one situation with exactly two possible outcomes: the transfer is completed or it is not completed.
2	Continuous	Uniform	The variable is bounded between two values, with no part of the interval expected to be more likely than another.
3	Discrete	Binomial	The variable counts the number of subscriptions among a fixed number of 80 comparable users, with each user having two possible outcomes.
4	Discrete	Poisson	The variable counts separate incidents occurring during a fixed period, with incidents occurring independently and without a predictable timetable.
5	Continuous	Triangular	The variable has a minimum, a maximum, and one most common value between them, with values becoming less common toward either extreme.
6	Discrete	Geometric	The variable counts the number of trials required until the first defective component is found.
7	Continuous	Linear	The variable is bounded and its probability density increases steadily as the value approaches the upper limit.
8	Discrete	Binomial	The variable counts the number of purchases among a fixed number of 120 comparable customers, with each customer having two possible outcomes.
9	Continuous	Normal	Values cluster around a central value, with approximately symmetric behaviour and increasingly low frequencies farther from the mean.
10	Discrete	Poisson	The variable counts separate cancellations occurring during a fixed time period, with events occurring independently and without a recurring timetable.



2 Delivery Time for a Local Grocery Order

[C2-4]

A Colombian grocery-delivery company studies the additional time required to deliver orders during a busy evening period. For a particular delivery zone, the added delivery time X can range from 0 to 10 minutes. A delay of any length within this interval is possible, and longer delays become steadily more common. Historical data indicate that a 10-minute delay is 4 times as common as no additional delay.

Questions/Tasks.

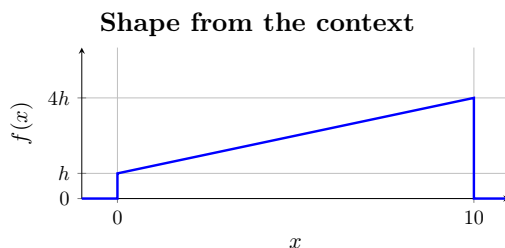
C2 Determine the complete probability density function for the added delivery time. Then determine the probability that the added delivery time exceeds 7 minutes, and the probability that it is between 3 and 7 minutes. [4]

C2 - Linear probability distribution

[4]

2.1 Support and Shape

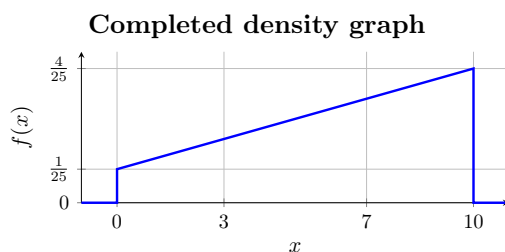
The random variable X has support $[0, 10]$. The context describes a steadily increasing straight-line density whose height at 10 is four times its height at 0. Write the endpoint heights as h and $4h$. The density is zero outside the support.



2.2 Total Area = 1

The total area under a probability density function must equal 1. Therefore, the trapezoid formed by the density has area 1:

$$1 = \frac{1}{2}(10)(h + 4h) = 25h, \quad h = \frac{1}{25}, \quad 4h = \frac{4}{25}.$$



2.3 Straight Line Rule

The density is a straight line passing through the two endpoint points

$$\left(0, \frac{1}{25}\right) \quad \text{and} \quad \left(10, \frac{4}{25}\right).$$

The slope is

$$m = \frac{\frac{4}{25} - \frac{1}{25}}{10} = \frac{3}{250},$$

and the y -intercept is

$$b = \frac{1}{25}.$$

Therefore, the probability density function is

$$f_X(x) = \frac{3x + 10}{250}$$

on the stated interval.

2.4 Complete pdf

Including the values outside the possible range of added delivery times, the complete probability density function is

$$f_X(x) = \begin{cases} \frac{3x+10}{250}, & 0 \leq x \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

2.5 Probability of More Than 7 Minutes

The company wants to know the chance that an order takes more than 7 additional minutes to deliver. This corresponds to the area under the density from 7 to the maximum possible delay of 10 minutes:

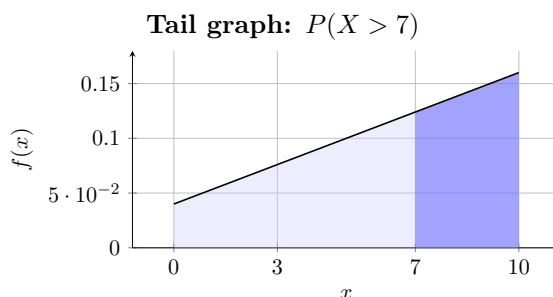
$$P(X > 7) = \frac{1}{2}(10 - 7)(f_X(7) + f_X(10)).$$

Since

$$f_X(7) = \frac{31}{250} \quad \text{and} \quad f_X(10) = \frac{4}{25},$$
$$P(X > 7) = \frac{1}{2}(3) \left(\frac{31}{250} + \frac{4}{25} \right) = \frac{33}{100}.$$

Thus, the probability of more than 7 minutes of additional delivery time is

$$P(X > 7) = 0.33.$$



2.6 Probability Between 3 and 7 Minutes

The company also wants to know the chance that the added delivery time falls between 3 and 7 minutes. This corresponds to the area under the density between these two times:

$$P(3 \leq X \leq 7) = \frac{1}{2}(7 - 3)(f_X(3) + f_X(7)).$$

Since

$$f_X(3) = \frac{19}{250} \quad \text{and} \quad f_X(7) = \frac{31}{250},$$
$$P(3 \leq X \leq 7) = \frac{1}{2}(4) \left(\frac{19}{250} + \frac{31}{250} \right) = \frac{2}{5}.$$

Thus, the probability of between 3 and 7 minutes of additional delivery time is

$$P(3 \leq X \leq 7) = 0.40.$$

